Skill Flow – Online Earning Platform

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) "Skill Flow – Online Earning Platform" was developed by "Muhammad Safi Ullah**, 21-Arid-471** "Fahad Iqbal Paracha**, 21-Arid-411" and** "Zarar Shahid**, 21-Arid-532"** under the supervision of "Dr. Sidra Tahir" and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Science.

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Executive Summary

The "SkillFlow" project is an innovative initiative designed to empower local communities, particularly women in small industries, by providing a dedicated platform for buying and selling non-technical skills. SkillFlow addresses the limitations of existing freelancing platforms, which predominantly focus on leaving individuals with local, non- technical skills underserved. By connecting buyers and sellers in a user-friendly digital marketplace, SkillFlow aims to unlock economic opportunities for skilled individuals and foster entrepreneurship at a grassroots level.

The project is driven by the need to create an inclusive and accessible platform tailored to the socio-economic realities of Pakistan. Traditional freelancing platforms such as Fiverr and Upwork are often inaccessible to non-technical service providers, especially women, due to their focus on professional services and a lack of localization. SkillFlow responds to this gap by targeting non-technical skills such as tailoring, baking, writing, and home- based crafts, ensuring that skilled individuals, regardless of technical expertise, have a platform to monetize their abilities.

SkillFlow embraces an **agile methodology**, allowing for iterative development and adaptability to evolving user needs and feedback. This ensures that the platform remains user-focused, with enhancements and features added progressively to meet the demands of both buyers and sellers.

The platform is developed using a modern technology stack, including React.js for the frontend, Node.js for the backend, and MongoDB for data management. This technology stack ensures scalability, security, and seamless user experience, even as the user base grows.

Through this comprehensive initiative, SkillFlow aspires to transform the freelancing landscape for non-technical skills, providing a platform that not only empowers individuals economically but also strengthens local economies. By enabling individuals to monetize their skills and connect with a wider audience, SkillFlow promotes financial independence,

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “Dr. Sidra Tahir” for personal supervision, advice, valuable guidance and completion of this project. We are deeply indebted to him for encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

Muhammad Safi Ullah | Fahad Iqbal Paracha | Zarar Shahid

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Chapter 1: Introduction

SkillFlow targets local businesses, small industries, allowing individuals to sell their skills and work on their own schedules. Our system provides a platform for people to earn money by directly communicating with one another and selling their skills. Buyers can search for sellers based on their skills, and sellers can customize their profiles to categorize their work, making it easier for buyers to find them. Additionally, buyers will receive recommendations for top-rated sellers based on their preferences.

1.1. Brief

SkillFlow is an innovative web-based platform designed to cater to local businesses and individuals, particularly women in small industries. The primary goal of SkillFlow is to provide an accessible platform where people can showcase their skills and connect directly with buyers. Unlike existing freelancing platforms that focus on technical services, SkillFlow prioritizes non-technical, skill-based services such as sewing, baking, and crafting. By enabling direct communication between buyers and sellers, the platform fosters community-based commerce and financial independence.

1.1.1. Key features include:

- Seller profile customization for showcasing skills.
- Buyer search and filtering options for easy service discovery.
- Real-time communication for efficient negotiations and queries.
- Secure payment gateways for seamless transactions.

SkillFlow is specifically designed to cater to Pakistani users, considering local needs, such as mobile-first design, regional payment systems (Easypaisa, JazzCash), and support for small-scale businesses.

1.2. Relevance to Course Modules

The development of SkillFlow integrates concepts and practices from various course modules studied during the degree program:

1.2.1. **Software Engineering**: SkillFlow applies the software development lifecycle (SDLC), particularly Agile methodology, to ensure iterative and incremental development.

1.2.2. **Database Management Systems:** MongoDB is used for handling user profiles, transactions, and service details, demonstrating proficiency in schema-less database design.

1.2.3. **Web Development:** The platform utilizes React.js for the frontend and Node.js for the backend, showcasing knowledge of modern web technologies.

By leveraging these modules, the project showcases the practical application of academic knowledge in building a real-world solution.

1.3. Project Background

In developing countries like Pakistan, freelancing platforms such as Fiverr and Upwork cater to technical and professional skills, leaving non-technical skill providers underserved. For instance, women skilled in tailoring, baking, or writing often lack the exposure or resources to compete on these global platforms. Moreover, cultural and mobility restrictions further limit their access to traditional employment opportunities.

SkillFlow was conceptualized to address these challenges by providing a localized platform tailored to non-technical skills. The platform connects buyers and sellers within the same region, promoting small businesses and local commerce. It focuses on empowering marginalized groups by creating an avenue for them to monetize their skills without needing extensive technical knowledge or resources. By offering services in familiar categories and integrating culturally relevant features, SkillFlow ensures inclusivity and relevance.

1.4. Literature Review

The literature review focused on understanding the gaps in existing freelancing platforms and the potential for a platform like SkillFlow. Platforms like Fiverr and Upwork have been extensively studied for their ability to connect freelancers with clients globally. However, these platforms primarily cater to technical services such as web development, graphic design, and content creation. Non-technical skills, which often cater to local overlooked.

Studies on women's empowerment through digital platforms highlight the importance of localized solutions. For instance, research shows that when women are given the tools to monetize their skills, their financial independence and household decision-making capabilities improve significantly. Additionally, user-centric design principles emphasize the need for platforms to be intuitive, especially in regions with limited digital literacy. Research on payment systems and online marketplaces also highlights the importance of secure transactions and transparent fee structures to build user trust.

By combining insights from these studies, SkillFlow addresses a niche yet significant gap in the market. It focuses on creating a simple, secure, and efficient platform that fosters skill monetization while promoting local commerce.

1.5. Analysis from Literature Review

The analysis derived from the literature review identified several critical points. First, there is a noticeable gap in the freelancing market for non-technical skills. While global platforms dominate the technical freelancing space, local talent in non-technical categories remains underrepresented. **SkillFlow** directly addresses this by prioritizing these services and creating a platform where buyers can discover them easily.

Second, existing research emphasizes the importance of localized solutions. For example, a platform that integrates regional payment systems and caters to the cultural and economic realities of its users is more likely to succeed in markets like Pakistan. Third, user-centric design is paramount. A platform that is easy to navigate and accessible across devices will significantly enhance user engagement**. SkillFlow** incorporates these findings by focusing on mobile-first design, responsive interfaces, and features tailored to local needs.

Finally, literature highlights the role of trust and transparency in sustaining online marketplaces, particularly in developing regions where digital adoption is still growing. Users are more likely to use a platform that ensures secure transactions, provides clear service terms, and supports dispute resolution mechanisms. In response to these findings, SkillFlow integrates secure payment gateways, transparent rating and review systems, and strong administrative controls to resolve conflicts effectively.

1.6. Methodology and Software Lifecycle for This Project

SkillFlow follows the **Agile methodology**, which allows for iterative development and constant user feedback. Agile is particularly suited for projects where user needs may evolve, ensuring the final product aligns with stakeholder expectations.

1.6.1. Planning and Requirement Gathering:

The team conducted initial surveys and research to understand the target audience's needs, particularly small business owners and women in local industries.

1.6.2. Design:

Wireframes and prototypes were developed using Figma to visualize the platform's structure and flow. Regular feedback from stakeholders ensured the design was user-centric.

1.6.3. Development:

The platform was built using React.js for the frontend and Node.js for the backend. MongoDB was used as the database to handle dynamic data efficiently.

1.6.4. Testing and Deployment:

Regular testing ensured that the platform was free of bugs and met performance standards. The final product was deployed with scalability in mind.

1.6.5. Rationale behind Selected Methodology

Agile was chosen for its flexibility and ability to adapt to changing requirements. Unlike traditional waterfall models, Agile allows the team to deliver functional prototypes quickly, gather feedback, and incorporate changes in subsequent iterations. This ensures the final product is not only functional but also aligned with user expectations. Agile's focus on collaboration and incremental development was crucial in creating a platform as dynamic as **SkillFlow.**

Agile also enhances risk management by breaking the project into manageable sprints where progress is continuously evaluated, allowing potential issues to be identified and resolved early rather than accumulating over time. This iterative approach keeps the development process highly transparent and encourages constant communication between developers, designers, and stakeholders, making it easier to prioritize features and adjust the scope as needed. By delivering value in small, usable increments.

Chapter 2: Problem Definition

In this chapter, we will discuss the challenges that the project, "Newborn Disease Prediction through Healthcare Data Analysis," aims to address. The main problem that the project seeks to solve is the limitations of traditional methods in detecting neonatal diseases. These methods are often associated with delays, potential errors, and an inability to efficiently handle vast amounts of healthcare data. The chapter will also outline the desired outcomes, which focus on developing a robust predictive model to enable early disease detection in newborns. By examining the existing gaps in neonatal healthcare and the constraints of manual approaches, this chapter sets the foundation for a detailed exploration of the project's problem domain and its expected impact on healthcare outcomes.

2.1. Problem Statement

In Pakistan, a significant portion of the population possesses valuable non-technical skills, yet they lack an accessible platform to monetize these talents. Traditional freelancing platforms prioritize technical services, leaving non-technical service providers, particularly women in small industries, underserved. Additionally, societal and cultural constraints often prevent these individuals from accessing traditional employment opportunities.

SkillFlow seeks to address these issues by providing a dedicated platform for non- technical skills. The platform connects buyers and sellers, allowing sellers to create profiles, showcase their work, and set prices. Buyers can easily search for services using filters such as category, location, and price. By integrating secure payment options and a user-friendly design, **SkillFlow** ensures a seamless and trustworthy experience for all users.

This project not only empowers individuals to achieve financial independence but also promotes local economic growth and the preservation of cultural skills. By addressing the specific needs of Pakistani users, **SkillFlow** stands as a transformative solution for fostering entrepreneurship and community engagement.

SkillFlow plays a crucial role in boosting digital entrepreneurship by giving Pakistani freelancers a dedicated platform to showcase their services, connect with clients, and earn sustainable income. By facilitating access to online work opportunities.

2.2 Deliverable and Development Requirements

In the context of our project **"SkillFlow – Freelancing Platform for Non-Technical Skills"**, the deliverables and development requirements are as follows:

2.2.1 Deliverables:

Deliverables are the tangible outputs that will be produced during the development of SkillFlow, serving as milestones to evaluate progress and ensure alignment with project goals.
Potential deliverables for our project could include:

2.2.1.1 Functional Software Application:

A fully functional web-based platform that enables freelancers and clients to register, post services, hire talent, communicate, and manage orders.

2.2.1.2 Database Infrastructure:

A secure and scalable database system to store user profiles, services, transactions, messages, reviews, and activity logs.

2.2.1.3 User Interface Designs:

Modern and user-friendly web designs for major platform modules including landing pages, dashboards, seller panels, buyer panels, and admin sections.

2.2.1.4 Documentation and User Guides:

Technical documentation covering system architecture, APIs, and setup instructions, along with user manuals and training guides for freelancers, clients, and administrators.

2.2.1.5 Validation and Testing Reports:

Comprehensive reports outlining results of unit testing, integration testing, usability testing, and performance testing of SkillFlow features.

2.2.1.6 Training and Support Materials:

Guides, walkthrough videos, FAQs, and troubleshooting resources to help new users onboard and utilize the SkillFlow platform effectively.

2.2.2 Development Requirements:

These development requirements define the functional and non-functional expectations that SkillFlow must meet during the development lifecycle in order to deliver a reliable, efficient, and secure freelancing platform tailored to local needs:

2.2.2.1 Scalability and Performance:

SkillFlow is expected to support a rapidly growing user base without compromising responsiveness or uptime. The system must be designed with a scalable architecture so it can efficiently manage thousands of simultaneous users, extensive service listings from sellers, and multiple ongoing client-seller interactions. Performance optimization techniques such as load balancing, database indexing, caching, and asynchronous processing should be adopted to ensure quick loading times and smooth real-time messaging. Achieving high performance standards will minimize latency, prevent platform bottlenecks, and result in a seamless user experience even during peak traffic periods.

2.2.2.2. Security and Privacy:

Incorporate robust security measures to protect sensitive healthcare data, including encryption protocols, access controls, and compliance with data privacy regulations such as HIPAA.

2.2.2.3. Interoperability:

Design the system to integrate seamlessly with existing healthcare infrastructure and data sources, enabling interoperability with electronic health records (EHRs), laboratory systems, and wearable devices.

2.2.2.4. User-Friendly Interface:

To ensure widespread adoption among freelancers and clients with varying technical skills, SkillFlow must offer an intuitive, visually appealing, and easily navigable user interface. The platform should be responsive, adapting fluidly across different screen

sizes and devices, particularly mobile phones, which are the primary internet access tools for many users in Pakistan. Simplified user flows, clear visual hierarchy, easy onboarding, and minimalistic layouts should be used so that even inexperienced users can quickly learn and utilize platform features such as posting services, placing orders, and communicating with other users. This design focus enhances usability and keeps user dropout rates low.

2.2.2.5 Compliance with Ethical Standards:

Adhere to ethical standards and guidelines governing the collection, storage, and use of healthcare data, ensuring transparency, fairness, and respect for patient rights throughout the development process.

Chapter 3: Requirement Analysis

This chapter delves into the meticulous analysis of requirements for the project, "Newborn Disease Prediction through Healthcare Data Analysis." Central to this analysis is the Software Requirements Specification (SRS) report, which meticulously outlines the essential functionalities, features, and specifications crucial for the successful development and implementation of the system. By detailing the project's scope, user needs, and system functionalities, the SRS report serves as a foundational document guiding subsequent project phases.

The chapter emphasizes the critical role of requirement gathering in shaping the project's trajectory. Through comprehensive analysis and stakeholder engagement, the project team ensures a thorough understanding of user requirements and expectations. This meticulous approach enables the identification of key use cases that drive the development of the system.

3.1. Use Cases

The use cases outlined in this section provide a detailed representation of user-system interactions, covering various scenarios and roles. They clarify system functionality and validate requirements by capturing these interactions. Additionally, they showcase the system's adaptability to diverse user needs, addressing tasks from data input to disease prediction and recommendation generation. By fostering effective communication between stakeholders and the development team, these use cases ensure alignment of requirements and expectations, guiding the development of a solution that efficiently meets user needs.

Use cases are used in a document to clearly describe how different users will interact with a system and what functionality the system must provide in order to satisfy business requirements. They present system behavior in simple, real-world scenarios so that both technical and non-technical stakeholders can easily understand what the system should do. By defining the actors involved and the sequence of steps they follow to achieve a specific goal, use cases help developers design appropriate features, assist testers in creating relevant test scenarios, and support project managers in prioritizing and validating requirements.

Following is the use case diagram according to the project SkillFlow.

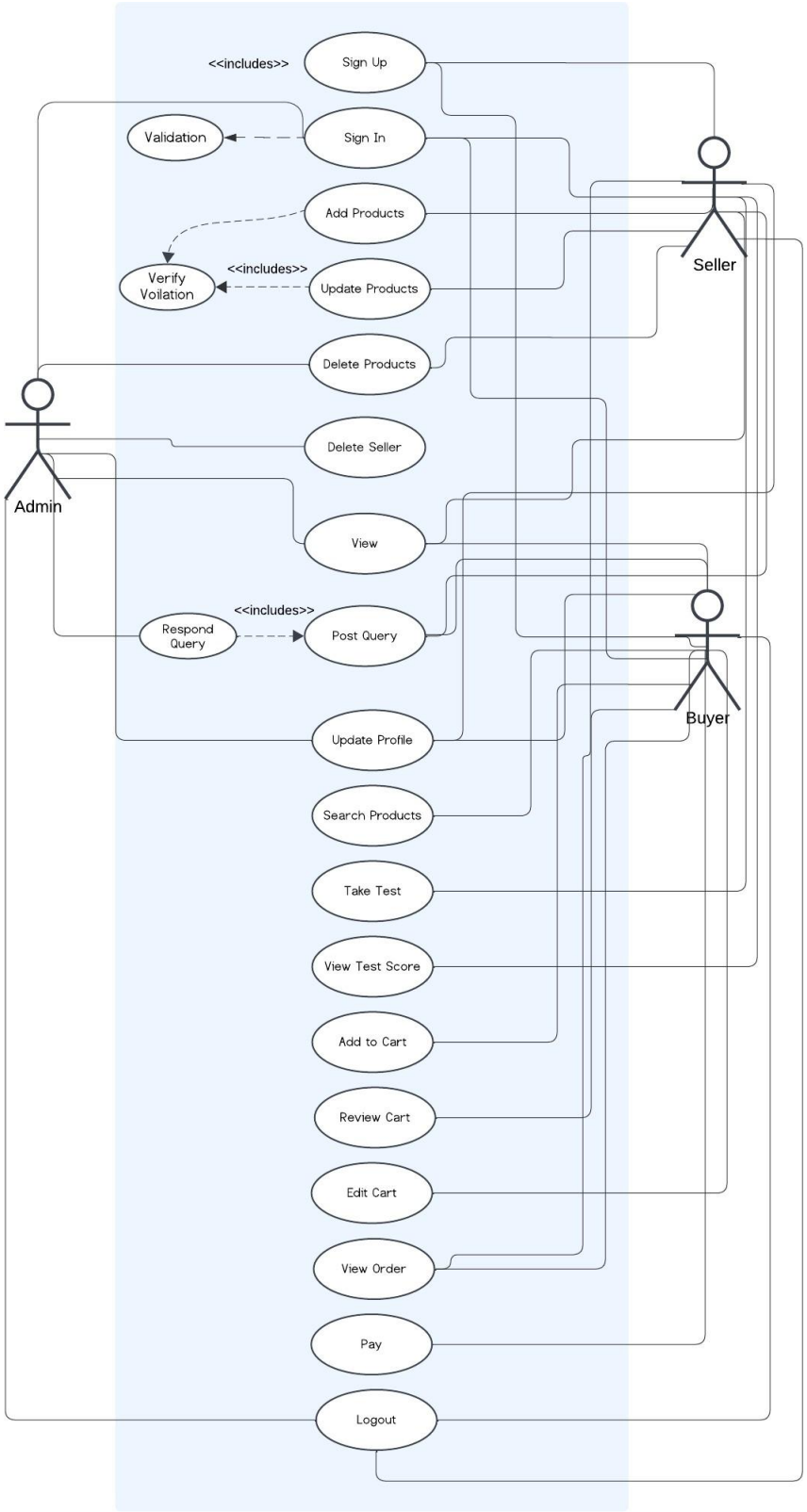

Figure 3.1: Use Case diagram

3.1.1. Actors Description:

3.1.1.1 User (Parent):

Description: The User (Parent) represents parents or guardians of newborns interacting with the system. Their primary role involves registering on the platform, providing necessary information about the newborn, and accessing predictions related to the newborn's health. They contribute essential data for analysis and receive personalized predictions.

3.1.1.2 Healthcare Professionals:

Description: Healthcare Professionals encompass various medical practitioners involved in newborn care, such as doctors, nurses, and specialists. Their role includes registering on the system, accessing newborn data, providing additional input for prediction analysis, and utilizing prediction results in clinical decision-making. They contribute valuable expertise to enhance prediction accuracy.

3.1.1.3 System Admin:

Description: The System Admin is responsible for overseeing the platform's functionality, ensuring data security, and managing user accounts. They have access to administrative tools for system maintenance, updates, and account management.

3.1.1.4 System:

Description: The System actor represents the entire software system itself. It serves as a collective term for the integrated components, including the user interface, database, prediction models, and administrative tools. Interactions with the System actor involve the flow of information, processing of data, and delivery of predictions to the relevant actors.

3.1.1.5 Database:

The database serves as the central repository that stores all critical information generated and used by the system, such as user profiles, transactions, service listings, messages, and activity logs. It enables efficient data retrieval by responding to queries from various system components whenever information is requested during operations.

3.2 Use Case Description:

3.2.1 Registration

Use Case ID:	ID-01
Use Case Name:	User Registration
Actors:	Buyer, Seller
Description:	This use case represents the process of a user registering on the system to gain access to healthcare data analysis and newborn disease prediction functionalities.
Trigger:	User decides to register on the system.
Preconditions:	The user has access to the system. The user is not already registered.
Post conditions:	The user is successfully registered on the system. The user can now log in and access other functionalities.
Normal Flow:	User navigates to the registration page. System presents a registration form. User fills in the required information (e.g., username, password, personal details). System validates the entered information. If validation is successful, the system registers the user. System notifies the user of successful registration.
Alternative Flows:	If the user enters invalid information (e.g., weak password, duplicate email), the system prompts the user to correct the details. If the user fails to confirm their email, registration remains incomplete.
Includes:	Email confirmation (if implemented)
Special Requirements:	The system should enforce strong password policies.
Assumptions:	Users have valid and unique email addresses.
Notes and Issues:	None

Table 3.2.1: Registration

3.3 Functional Requirements

The specific functionalities and features that the SkillFlow must possess to fulfill user needs and achieve project objectives. The proposed system's functional requirements are outlined as follows:

3.3.1 Secure User Registration and Login:

- Ensures the safety of user data through encrypted storage and transmission.
- Implements strong password policies to minimize the risk of unauthorized access.
- Offers additional security measures, such as email verification and two-factor authentication, for identity validation.
- Limits the number of login attempts to protect against brute-force attacks.
- Provides secure session management, ensuring user accounts remain protected during active sessions.

3.3.2 Profile Customization and Service Listing:

- Sellers can create detailed profiles, including personal descriptions, skill categories, and portfolio images or videos.
- Profiles allow categorization of services (e.g tailoring, baking) for better discoverability.
- Sellers can update their profiles dynamically to reflect new skills or updated service offerings.
- Highlighted sellers' expertise, building trust with potential buyers through detailed and professional presentations.

3.3.3 User Registration and Authentication:

- Users, including healthcare professionals and parents, can register on the system to access its functionalities.
- Robust user authentication mechanisms will be implemented to ensure secure and authorized access.

- Role-based access control will restrict system features based on user type to prevent unauthorized actions.
- Allows users to update their profile information and manage account settings through a personalized dashboard.

3.3.4 User Interface (UI):

- The system will feature a user-friendly interface for healthcare professionals to input patient data, potentially incorporating visualizations to aid in comprehension of predictions and contributing factors.

3.3.5 Feedback and Information Input:

- Healthcare professionals and parents will have access to a user-friendly interface to input relevant information about newborns.
- Feedback forms and surveys will be available for users to provide additional insights and observations.

3.3.6 Admin Functionality:

- Administrative users will possess privileges to manage different user accounts, ensuring proper access control.
- The admin interface will facilitate monitoring and maintenance of the system, including updating algorithms and managing data sources, while adhering to relevant data protection regulations.

3.3.7 Financial Transactions Handling:

- Integrates with local payment gateways like Easypaisa and JazzCash, as well as international options like PayPal, for user convenience.
- Processes payments securely using encrypted channels to protect sensitive financial information.
- Implements escrow-like systems, holding payments until services are successfully delivered and confirmed by buyers.

- Tracks and logs transactions to ensure transparency and provide records for users and the platform's admin team.
- Builds trust by ensuring fair and secure payment processing for both buyers and sellers.
- Supports real-time payment status updates to keep both parties informed throughout the transaction lifecycle.
- Automatically calculates platform service charges and commissions before releasing funds to sellers.

3.4 Non-Functional Requirements

The SkillFlow's non-functional requirements encompass crucial aspects of usability, reliability, performance, supportability, and compliance with regulatory standards. They are outlined as follows:

3.4.1 Performance:

- The system must handle high user concurrency, allowing multiple users to interact with the platform simultaneously without experiencing delays or lag.
- Ensures fast response times for critical operations such as logging in, searching for services, and processing transactions.

3.4.2 Reliability:

- Ensures the platform operates consistently under different conditions and provides uninterrupted services to users.
- Implements failover mechanisms and backups to minimize downtime during server failures or maintenance.

3.4.3 Scalability:

- Focuses on creating an intuitive and user-friendly interface that caters to individuals with minimal technical skills.

- Provides clear navigation, well-structured layouts, and helpful tooltips to guide users through various features.

3.4.5 Supportability:

- The system should facilitate easy troubleshooting and issue resolution for users and administrators alike.
- Adequate user support channels, including FAQs and user guides, should be accessible to aid users in navigating the system effectively.

3.4.6 Usability:

- The system's design must adhere to healthcare data privacy regulations and standards to ensure the confidentiality and security of patient information.

Chapter 4: Design and Architecture

This chapter explores the design and architecture of the "SkillFlow" system, emphasizing key principles and decisions vital for project success. It outlines the significance of thoughtful design considerations in meeting project objectives efficiently. From conceptualizing system structure to detailing architectural components, it serves as a blueprint for translating requirements into a functional solution. Providing insights into the project's technical foundation, it fosters a deeper understanding of the design process.

4.1. System Architecture

As system design varies from system to system, the user needs to have an architectural view of the whole system. The system architecture defines the architecture of the entire system.

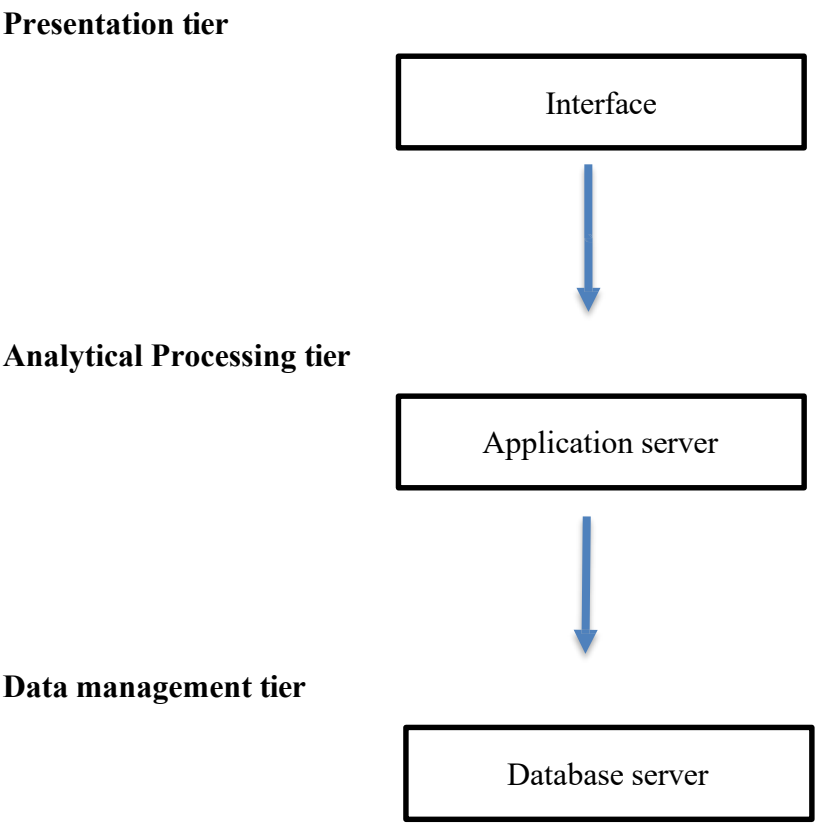

Fig 4.1: System Architecture

4.1.1 Presentation Tier (Interface):

- This tier handles the user interface where buyers and sellers interact with the SkillFlow platform. Buyers can browse services, search for specific skills, and communicate with sellers, while sellers can create profiles, list their services, and manage their accounts.
- A clear separation of the presentation layer is implemented to provide a **user-friendly interface**. Since many users in Pakistan may have limited technical skills, the interface is designed to be intuitive, responsive, and mobile-friendly, ensuring accessibility across devices.
- Technologies used for this layer include **React.js** for developing a dynamic and interactive UI and **Figma** for prototyping and user experience design.

4.1.2 Analytical Processing Tier (Application Server):

- The application server, or the business logic tier, processes the core functionalities of the platform, such as managing user authentication, profile creation and messaging systems.
- This layer also handles recommendations, where personalized suggestions are generated for buyers based on their search history, preferences, and ratings.
- Technologies like **Node.js** (backend framework) and **Express.js** (middleware) are used to manage the application logic efficiently.

4.1.3 Data Management Tier (Database Server):

- The data management tier is responsible for securely storing and retrieving platform data, including user profiles, service listings, messages, and transaction records.
- By isolating the data layer, the platform ensures **efficient data management** and **enhanced security**, particularly for sensitive information such as payment details and user credentials.

4.2. System Design

Systems design is the process of defining elements of a system like components, modules, architecture, and their interfaces and data for a system based on the specified requirement.

4.2.1 UML Behavioral Diagrams

Structural UML diagrams show how the system is structured, including the classes, objects, packages, components, etc. in the system and the relationships between the elements.

4.2.1.1 Component Diagram:

This diagram illustrates the system's components and their interactions, offering a visual representation of the system's architecture. It showcases how various components collaborate to achieve system functionality and highlights dependencies between them.

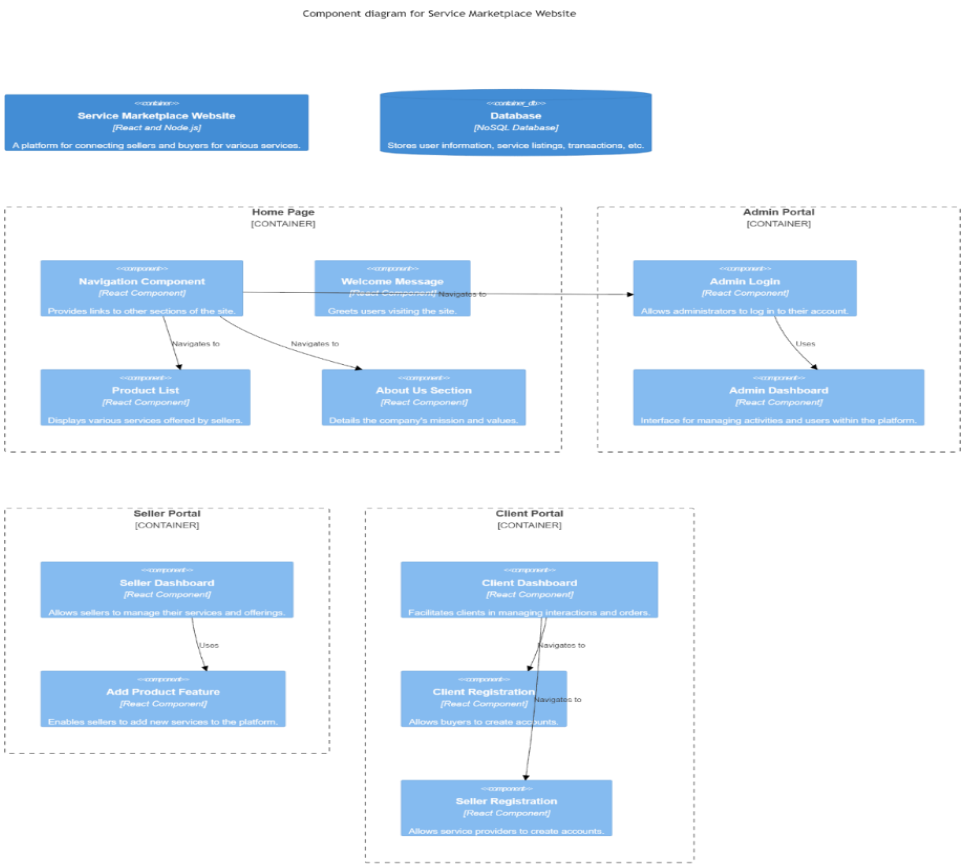

Fig 4.2.1.1: Component Diagram

4.2.1.2 Deployment Diagram:

This diagram depicts the physical deployment of software components across hardware nodes, illustrating how the system's modules are distributed and interconnected within a network infrastructure. It visualizes the deployment architecture, including servers, databases, and other resources, facilitating effective system deployment and management.

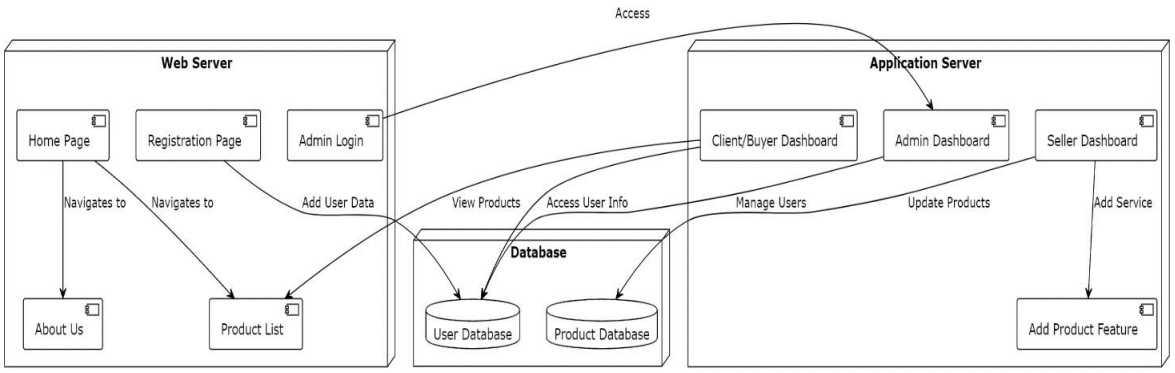

Fig 4.2.1.2: Deployment Diagram

4.2.1.3 UML Behavioral Diagrams

UML Behavioral Diagrams offer a clear visual representation of dynamic system behavior, interactions among components and entities, and task accomplishment. They aid in comprehending functionality, message passing, and state transitions, promoting efficient system design and stakeholder communication.

4.2.1.4 Activity Diagram for Buyer:

The flow of activities within a system, illustrating the sequence of actions or steps required to accomplish a specific task or behavior.

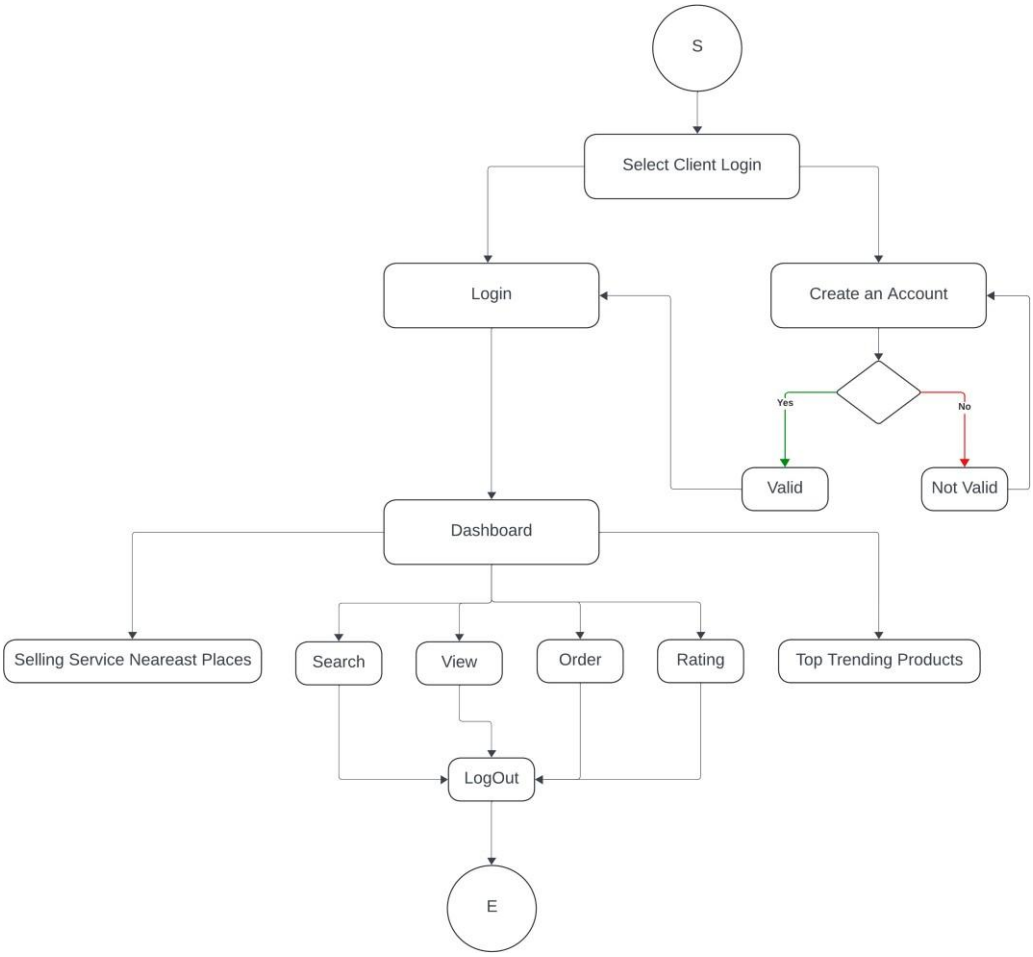

Fig 4.2.1.4: Activity Diagram

4.2.1.5 Activity Diagrams for Seller:

This diagram models the behavior of individual components or objects within the system, depicting their states and transitions between them. It offers a visual representation of how the system responds to events and stimuli, guiding the understanding of its dynamic behavior.

The Activity Diagram for the seller outlines the flow of activities that occur as a seller interacts with the SkillFlow platform, such as signing in, creating service listings, managing orders, and responding to client requests. By representing each action as a node and showing the transitions between them based on events or decisions, the diagram illustrates the logical sequence of tasks and possible alternative paths. This visual flow helps stakeholders understand the seller's operational process, identify dependencies between actions, and analyze how the system supports different seller behaviors from start to end.

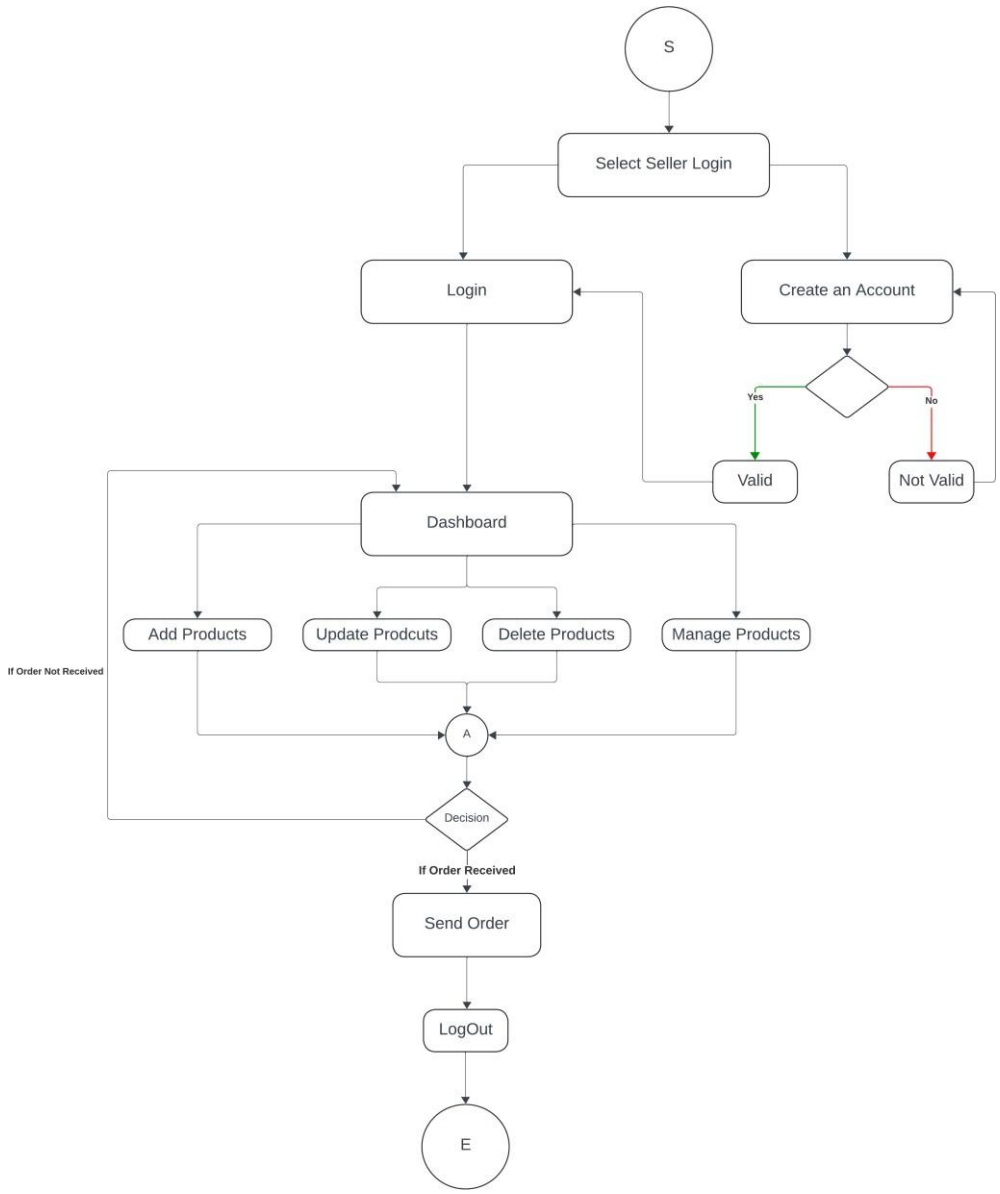

Fig 4.2.1.5: State Machine Diagram

4.2.2 UML Interaction-diagrams

UML Interaction Diagrams are used to model the dynamic behavior of a system by focusing on how different objects communicate with each other during execution. They emphasize the exchange of messages and the order in which interactions occur, giving a clear view of system behavior over time. Sequence Diagrams, a primary type of interaction diagram, present this communication in a time-ordered manner by visually mapping the flow of messages between objects from top to bottom.

4.2.2.1 Sequence Diagrams

Sequence diagrams in UML illustrate the interactions between objects or components in a system over time, depicting the flow of messages between them.

4.2.2.1.1 Registration sequence Diagram

The Registration Sequence Diagram shows the sequential communication that occurs when a new user signs up on the SkillFlow platform. It captures how the user enters their details into the registration form, how the system sends this information to the authentication service, checks it against existing records in the database, and then creates a new user profile if the data is valid and unique. This diagram also depicts the confirmation response sent back to the user, signaling successful account creation or highlighting errors (such as duplicate email or invalid information). By mapping this interaction step-by-step, the diagram helps developers ensure a smooth and secure onboarding process that collects necessary user data while preventing unauthorized or duplicate accounts.

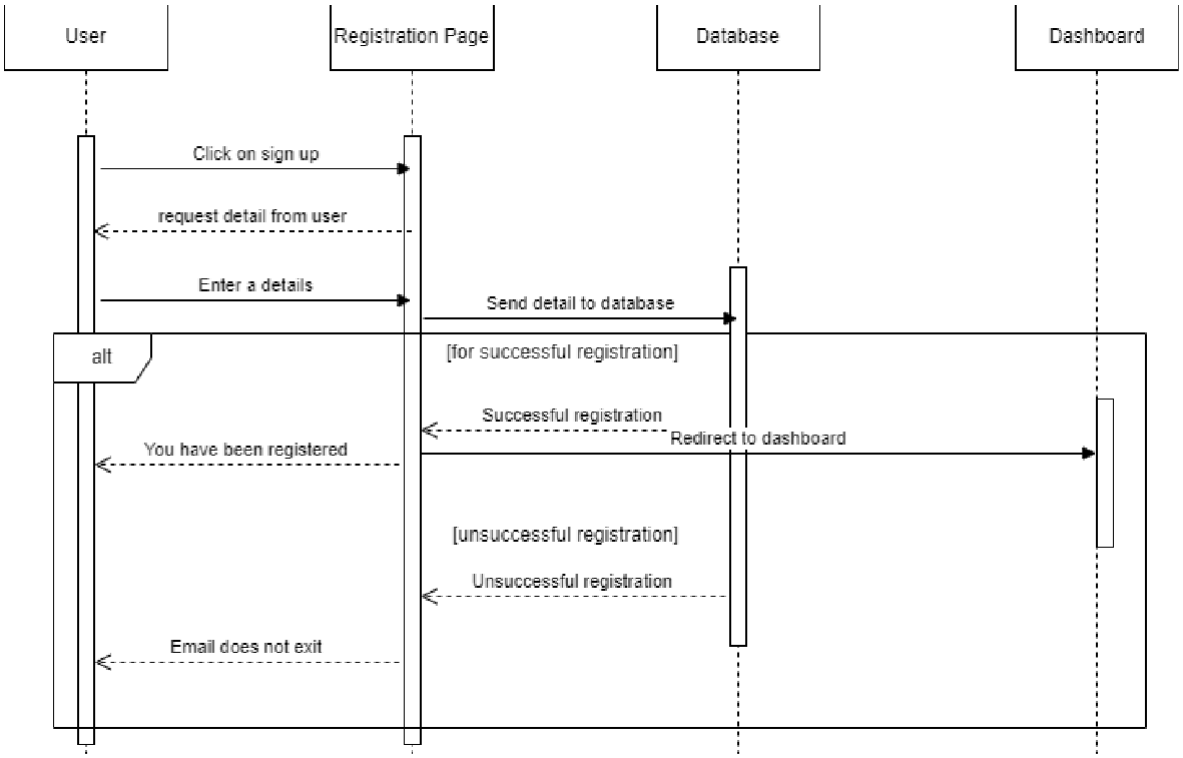

Fig 4.2.2.1.1: Register Sequence Diagram

4.2.2.1.2 Login Sequence Diagram

The Login Sequence Diagram represents the chronological interaction between the user and system components during the authentication process on SkillFlow. It outlines how a user submits login credentials, how the system forwards this information to the authentication service, checks it against stored data in the database, and then returns a success or failure response. By detailing this back-and-forth flow of messages, the diagram clarifies each step involved in validating user identity, managing sessions, and granting secure access to the platform.

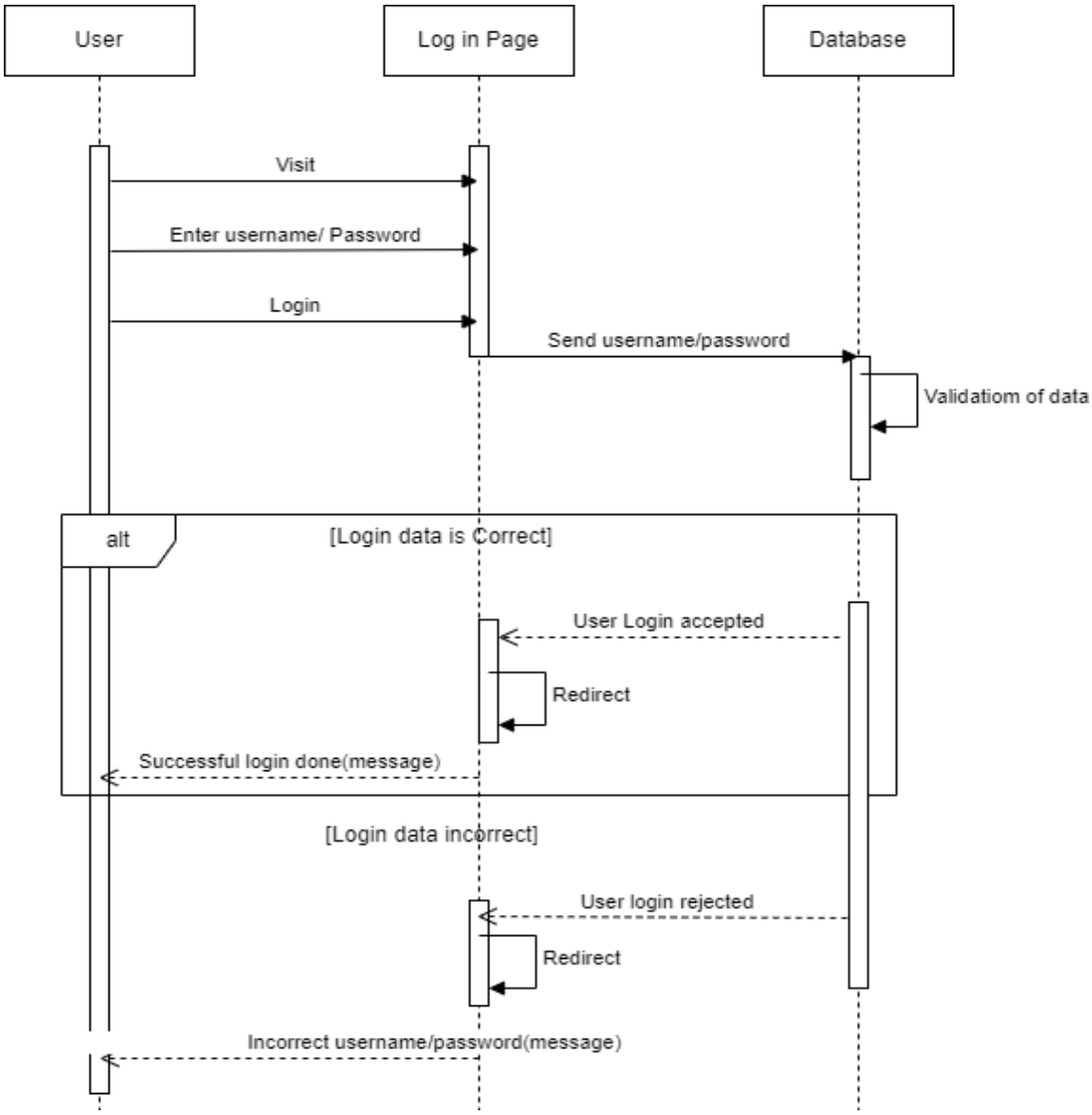

Fig 4.2.2.1.2: Login sequence diagram

4.2.2.1.3 Seller Sequence Diagram

The Seller Sequence Diagram illustrates the step-by-step interaction between the seller and various system components during key processes such as account creation, service posting, and order management. It visually maps out how the seller sends requests (like uploading a gig or responding to a client's inquiry) and how the system responds through different modules such as authentication, database storage, notification service, and order tracking. By showing the flow of messages and the sequence in which operations occur, this diagram helps developers and stakeholders understand the seller's journey in a logical, time-ordered manner. It ensures that all necessary system interactions are captured to support smooth seller operations on the SkillFlow platform.

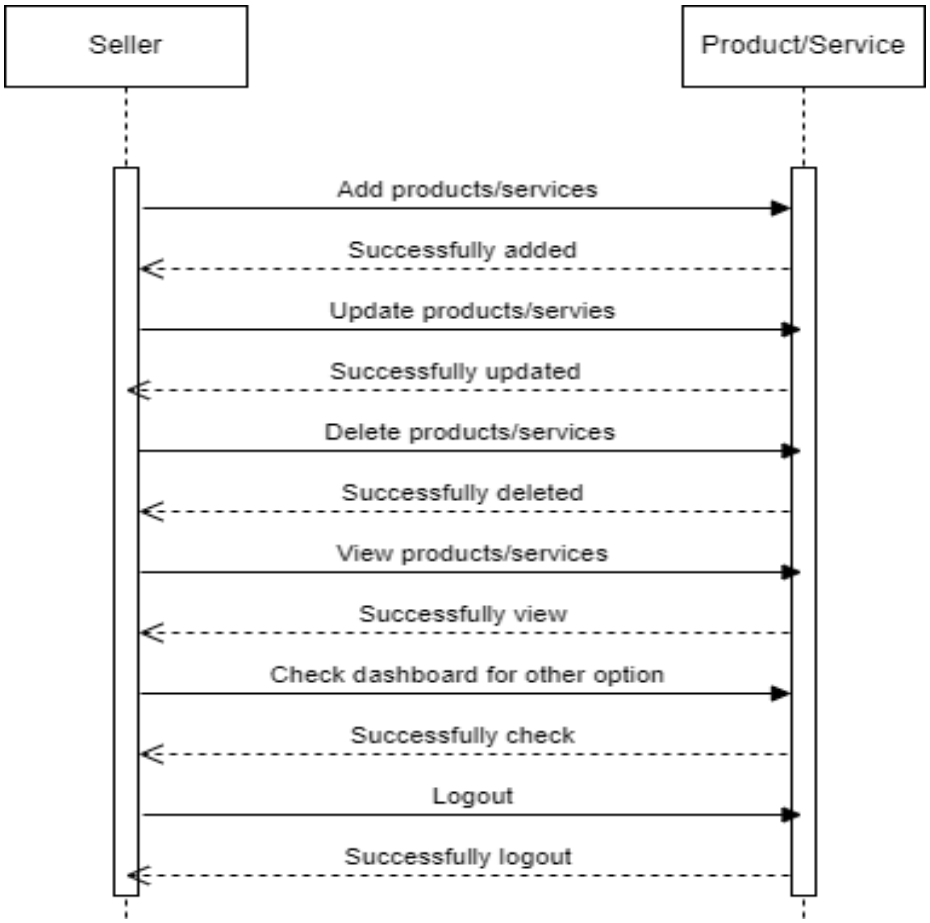

Fig 4.2.2.1.3: Seller Sequence Diagram

4.2.2.1.4 Class Diagram

Fig 4.2.2.1.4: Class Diagram

Chapter 5: Implementation

This chapter provides a comprehensive walkthrough of the implementation phase of the SkillFlow system, detailing how the designs and architecture outlined in Chapter 4 were translated into a fully functional web platform. It covers both frontend and backend development, describes the tools and technologies used, explains the database structure, and highlights key modules such as service discovery, booking workflows, real-time messaging, and profile management. The implementation adhered to modern software development principles, using React with Tailwind CSS and Redux Toolkit for the frontend, and Node.js with Express for backend logic. MongoDB Atlas serves as the primary database, with well-designed collections for users, services, transactions, and conversations. Axios handles client-server communication, while Socket.IO powers live chat and in-app notifications. The chapter begins with setting up the development environment—project scaffolding, dependency management, and CI/CD configuration— then walks through frontend component development, REST and WebSocket API design, middleware setup, and database schema modeling. It concludes by discussing integration strategies, automated testing approaches, and the challenges encountered—such as securing JWT flows and optimizing query performance—and the solutions applied. Each module was implemented in line with the functional and non-functional requirements, resulting in a maintainable, scalable, and secure platform ready for real-world deployment.

5.1. Component Diagram

The component diagram for SkillFlow presents a high-level visual overview of the system's core modules and their interactions. It divides the platform into distinct components—namely, the User Component, Driver Component, Admin Component, Database Component, Notification Component, and Middleware Layer. The User Component manages client and freelancer interfaces for service discovery, profile management, booking, and real-time chat. The Database Component persists user profiles, service listings, transactions, and message histories in MongoDB, supporting efficient queries through carefully defined indexes. The Notification Component handles in-app alerts, email notifications, and SMS reminders based on user preferences. The Admin Component provides

dashboards for user and content moderation, analytics, and dispute resolution, enforcing role-based access controls and audit logging. Sitting between client requests and business logic, the Middleware Layer applies security headers, CORS policies, authentication guards, input validation, and centralized error handling. By mapping out these components and their data flows—API calls, WebSocket events, and background job queues, the diagram clarifies dependencies and interfaces, promotes modular development, and makes it easier to update or scale individual parts without impacting the whole system. This holistic view is essential for guiding implementation, ensuring maintainability, and delivering a seamless experience.

Fig 5.1: Component Diagram

5.2. User Component:

The User Component serves as the face of SkillFlow, covering every page and interaction that a client or freelancer experiences. It includes screens for signing up and signing in, multi-step profile setup and editing, intuitive dashboards for discovering and managing services, and seamless messaging interfaces for real-time conversation. Under the hood, it relies on secure, token-based authentication to protect user sessions and form-level validation to prevent bad input. Responsive design techniques adapt the layout and content for phones, tablets, and desktops, while clear inline error messages and loading indicators guide users through each workflow. All network requests flow through a shared service layer that standardizes how APIs are called—automatically retrying failed requests, surfacing errors in a consistent way, and caching common data to reduce latency. Throughout, accessibility features such as keyboard-friendly navigation, ARIA-annotated components, and carefully chosen contrast ratios ensure the interface is usable by everyone. Performance enhancements—like dynamically loading only the code needed for each page, compressing or deferring large assets, and splitting bundles to keep initial downloads small—combine to deliver a fluid, responsive experience from the first click to the last.

5.2.1 Authentication Screen: Enables users to sign up, log in, and recover passwords.

5.2.2 Profile Dashboard: Allows clients and freelancers to view and edit their personal and service details.

5.2.3 Service Discovery Page: Provides search, filters, and sorting to find relevant services.

5.2.4 Service Listing Form: Lets freelancers create and publish new service offers.

5.2.5 Booking Workflow: Guides users through selecting a time slot, confirming details, and initiating payment.

5.2.6 Real-Time Chat Interface: Facilitates live messaging between clients and service providers.

5.2.7 Notifications Panel: Displays in-app alerts for messages, bookings, and status updates.

5.2.8 Reviews & Ratings Section: Enables users to submit and view feedback on completed services.

5.2.9 Settings & Preferences: Lets users configure account settings, notification preferences, and privacy options.

5.2.10 Help & Support Center: Provides FAQs, contact forms, and troubleshooting guides.

5.3. Admin Component:

The Admin Component delivers a unified dashboard for SkillFlow's administrators and moderators. It features user management screens for viewing, approving, or suspending client and freelancer accounts; service moderation panels for reviewing and controlling service listings; and transaction oversight tools for monitoring payments, refunds, and disputes. Role-based access controls ensure that only authorized staff can access sensitive functions, and every admin action— from account suspension to policy updates—is recorded in an audit log. The interface is designed for efficiency, with bulk-action controls, searchable tables and filters, and visual summaries such as charts of new signups, revenue trends, and the distribution of service categories. Behind the scenes, a set of secure REST APIs applies the same permission checks and data validations, guaranteeing that the dashboard both reflects and enforces the platform's integrity and governance policies.

5.3.1 User Management Panel:

View, approve, suspend, or delete client and freelancer accounts.

5.3.2 Dashboard:

Review, approve, or flag service listings for policy compliance.

5.3.3 Audit Log Viewer:

Track and review records of all admin actions for security and compliance.

5.3.4 Role & Permission Manager:

Configure and assign role-based access controls for different admin functions.

5.3.5 Content Management Interface:

Create, edit, and remove static pages, FAQs, and policy documents.

5.3.6 Notification Configuration:

Set up and manage global email/SMS templates and alert rules.

5.3.7 System Settings Panel:

Adjust platform-wide settings such as fees, time zones, and localization options.

This Admin Component is essential for keeping SkillFlow, secure, and responsive, enabling continuous platform improvement and reliability.

By providing authorized personnel with powerful oversight and control features, the Admin Component ensures that platform rules are enforced, inappropriate content

5.4 Database Component:

The database component covers the design, structure, and deployment of all persistent storage. This includes user accounts, service listings, transactions, messages, and notification records. A scalable document-oriented database is used for most application data, with carefully designed indexes (e.g. compound and geospatial) to support common queries efficiently. Relational or in-memory stores may be employed for sessions or coaching.

5.4.1 Users Collection:

Stores client and freelancer profiles, authentication credentials, roles, and preferences.

5.4.2 Services Collection:

It contains service listings with details such as title, description, price, category.

5.4.3 Transactions Collection:

Records booking details, payment statuses information.

5.4.4 Notifications Collection:

Logs in-app alerts, email and SMS notifications with user preferences and delivery status.

5.4.5 Ratings & Reviews Collection:

Stores feedback entries, star ratings, comments, and associations to users and services.

5.4.6 Sessions Collection:

Manages active user sessions, **refreshing** tokens, and expiration data for security and analytics.

5.5 Notification Component:

Notifications deliver timely updates via in-app alerts, emails, and SMS (or push) messages. A central notification service routes events—such as new messages, booking confirmations, or payment updates—to the appropriate channels based on user preferences. It supports batching for high-volume events, throttling to prevent spamming, and retry mechanisms for transient failures to the appropriate channels based on user preferences.

5.6 Choice of middleware:

Middleware functions act as the "pipeline" through which every request and response flows, allowing you to slot in shared logic without cluttering your core application code. At a high level, standard middleware takes care of universal concerns—things like setting security headers, parsing incoming payloads, handling cross-origin requests, logging traffic, and throttling excessive use. Alongside these, custom middleware can enforce authentication by checking user tokens and loading their context, validate inputs against predefined schemas, and catch errors in a centralized place to ensure consistent messaging and status codes. By organizing these layers thoughtfully, you keep your route handlers free to focus purely on the business tasks—whether that's creating a new booking, updating a profile, or sending a notification—while the middleware stack uniformly manages the common, cross-cutting aspects of every interaction.

- Authentication
- Logging
- Error handling
- CORS configuration
- Validating user input

5.6.1 User Interface:

The user interface (UI) is the visual and interactive part of a system through which users communicate with software or hardware. It includes all the elements that a user sees and interacts with—such as buttons, menus, icons, input fields, forms, and screens. A good UI is designed to be intuitive, visually appealing, and easy to use, enabling users to perform tasks efficiently without confusion. It translates the system's functionality into a clear, understandable format so users can navigate and control the system smoothly.

An effective user interface plays a critical role in the success of any application because it directly affects the user experience. If the UI is cluttered, confusing, or difficult to use, users

may become frustrated and abandon the system. On the other hand, a well-designed UI improves usability, reduces errors, and increases user satisfaction by guiding users naturally through the workflow. It also helps accommodate users with different levels of technical skill by simplifying complex processes into easy steps, thereby making the system more accessible and productive.

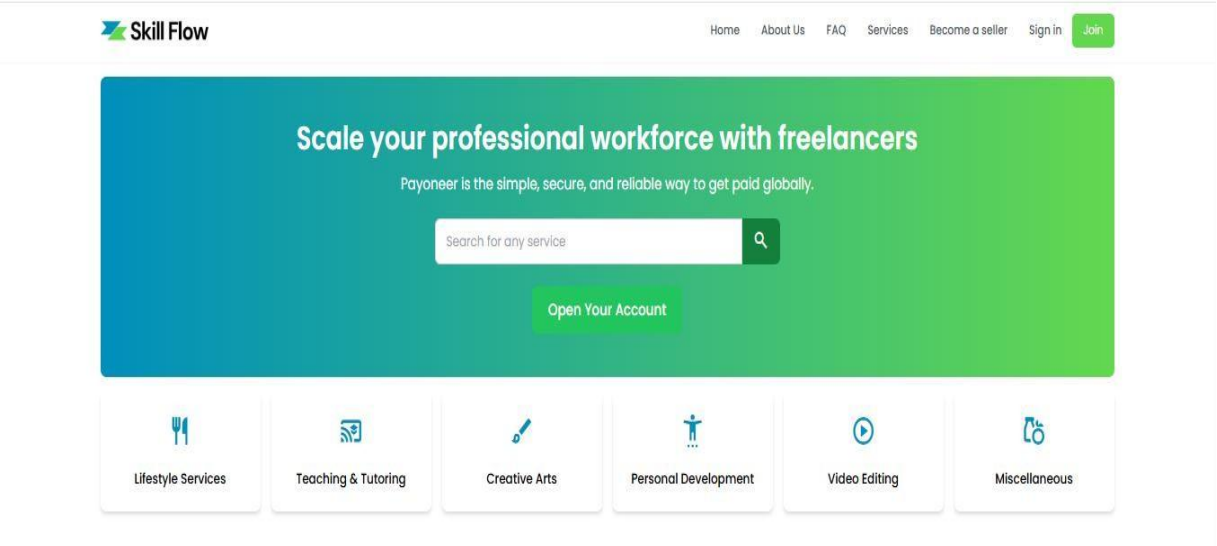

Fig 5.6.1: Home Page

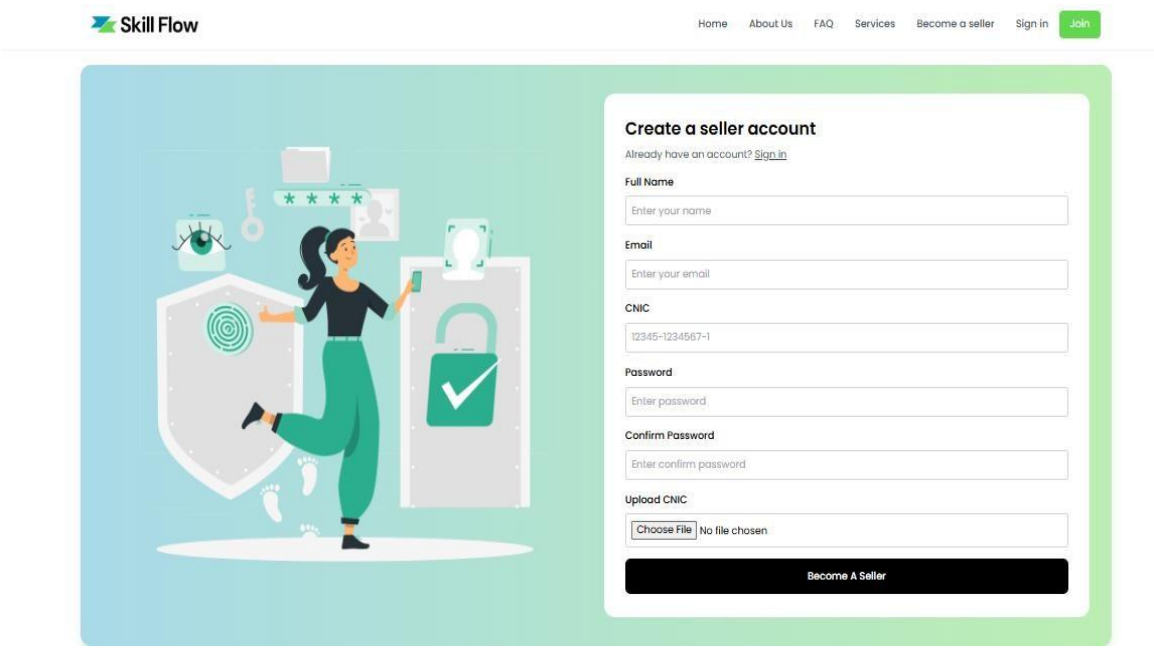

Fig 5.6.2: Seller Registration

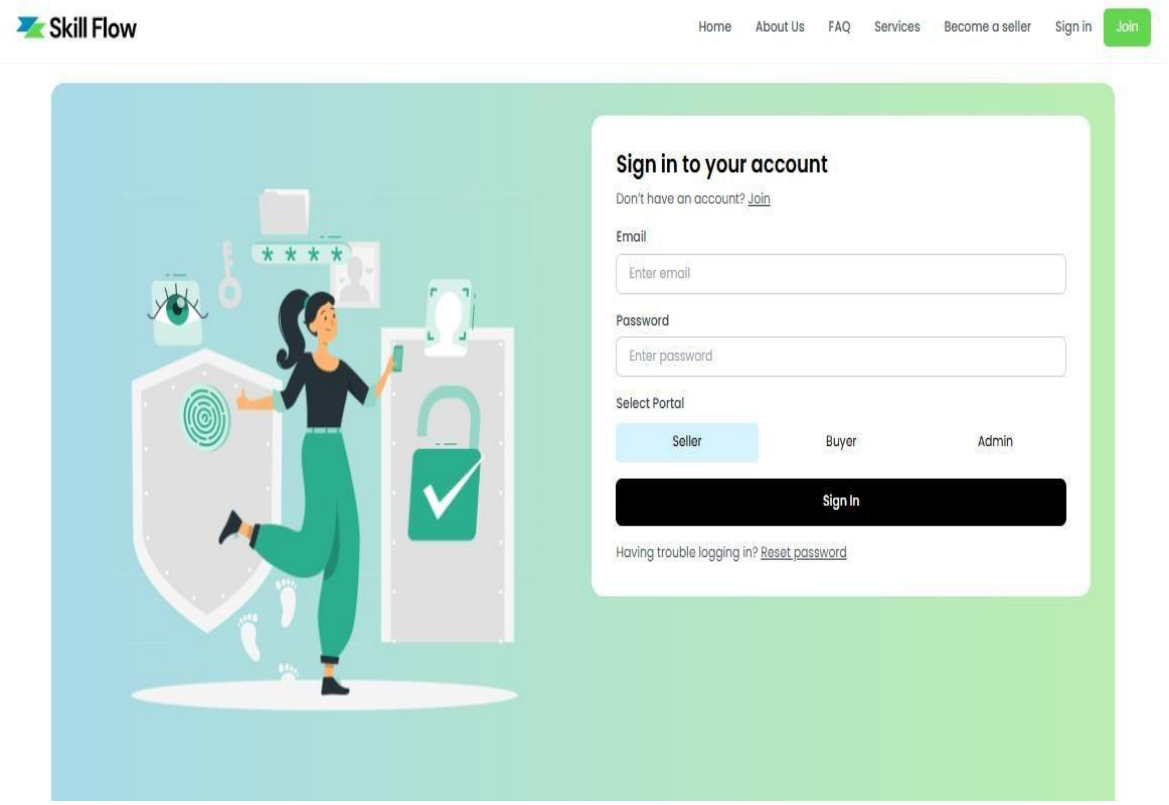

Fig 5.6.3: User Sign-In

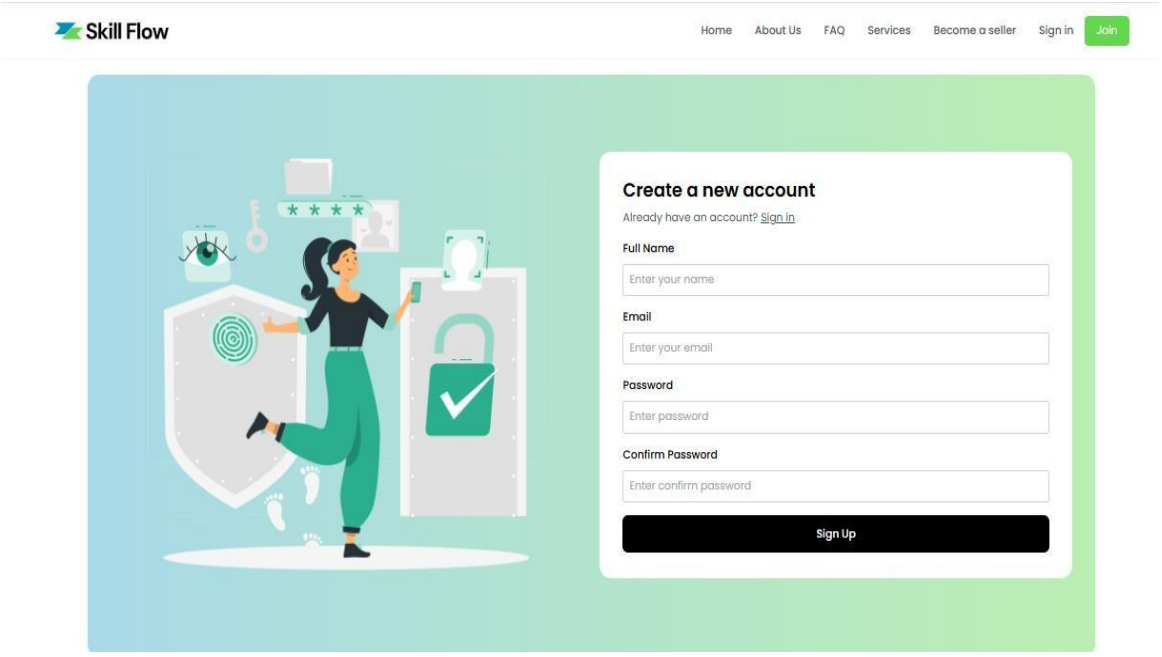

Fig 5.6.4: Buyer Registration

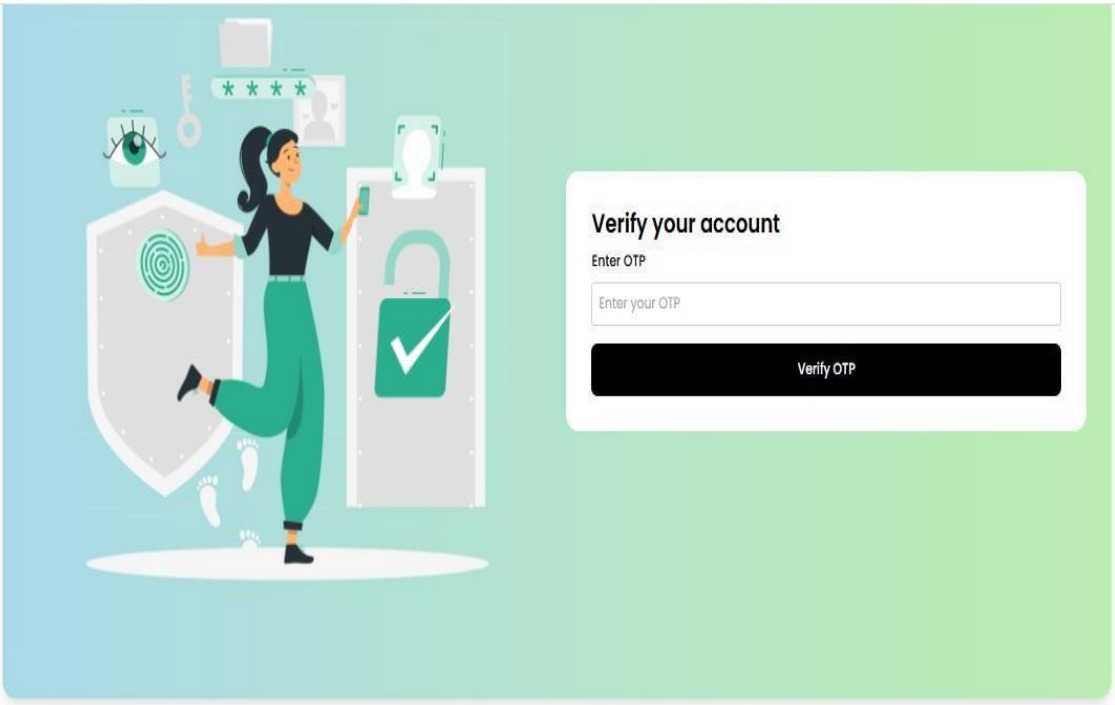

Fig 5.6.5: Verify OTP

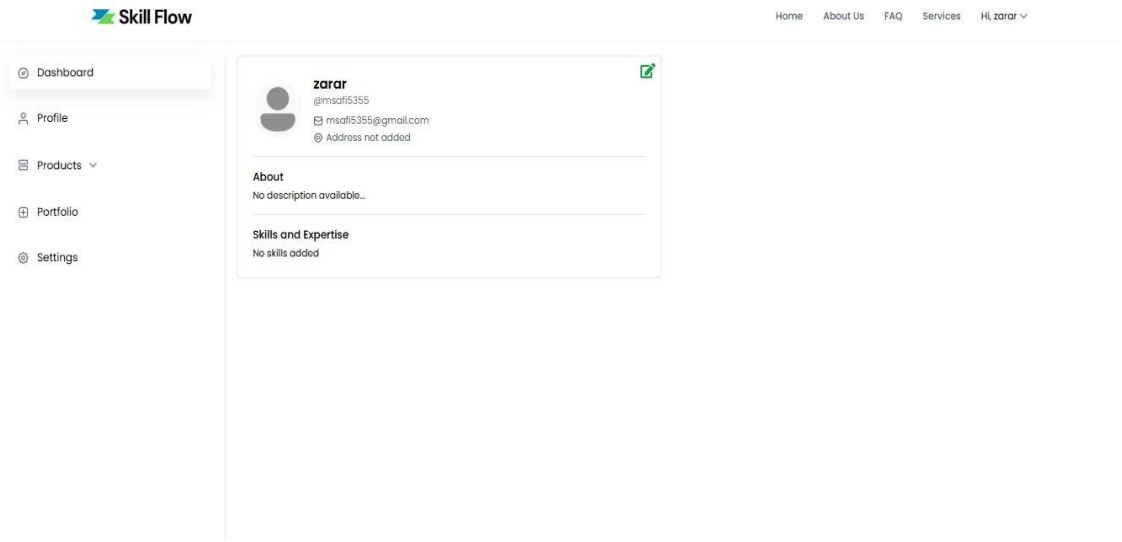

Fig 5.6.6: Dashboard - View

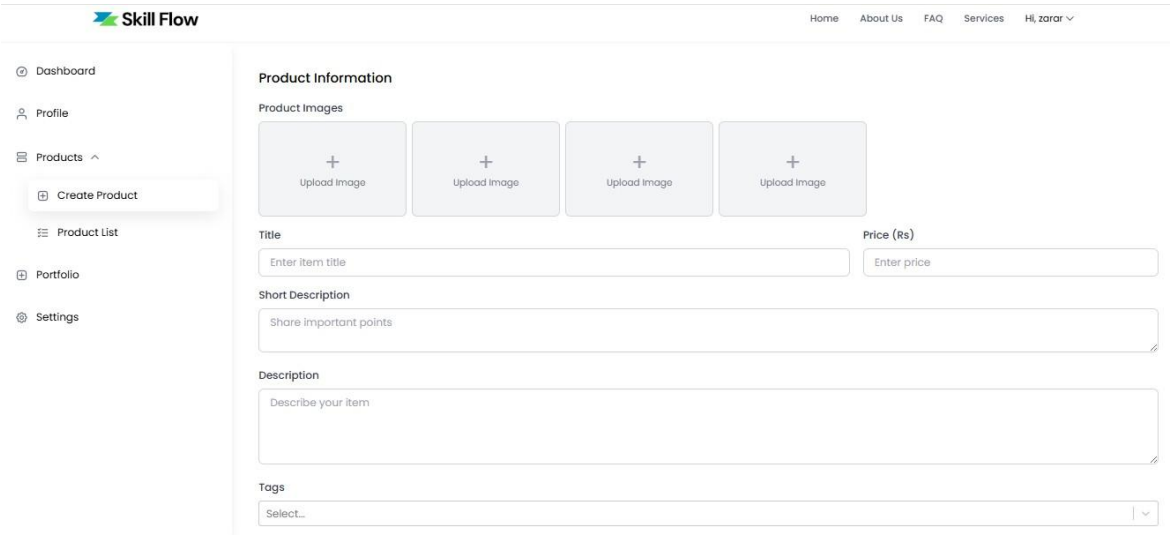

Fig 5.6.8: Create Product View

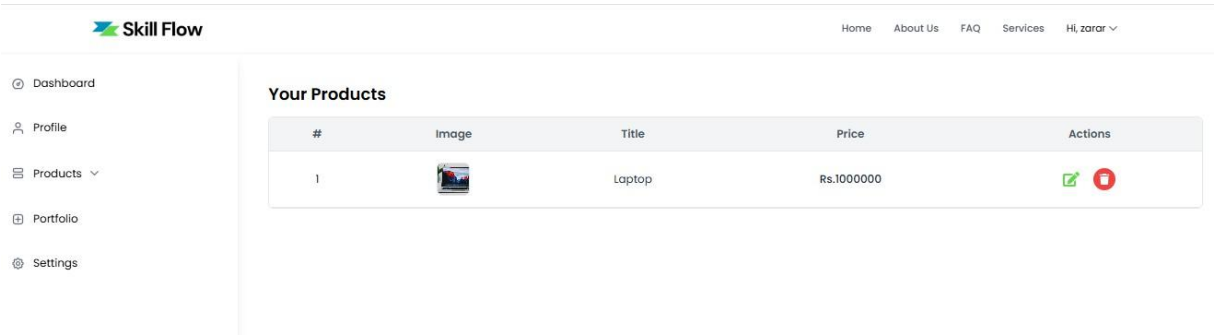

Fig 5.6.9: Product List View

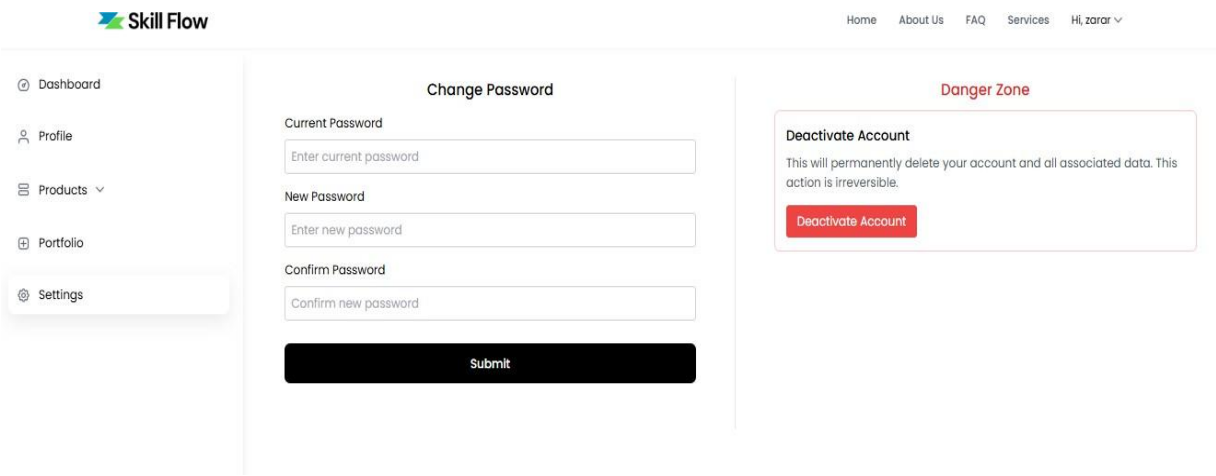

Fig 5.6.10: Setting View

Chapter 6: Testing and Evaluation

This chapter presents the comprehensive testing phase conducted for the SkillFlow platform to ensure the application functions correctly, securely, and efficiently under various conditions. It details the types of tests applied—functional testing, unit testing, integration testing, verification, performance testing, validation, and user acceptance testing (UAT). The purpose is to document how the team validated that the system meets the functional and non-functional requirements defined earlier. Testing was carried out across all core modules, covering frontend components (service discovery, booking flows, messaging UI), backend APIs (authentication, transactions, notifications), database operations (CRUD on users, services, conversations), and third-party integrations (payment gateways, email/SMS services, file storage).

6.1. Verification

Verification focuses on ensuring that each individual part of SkillFlow functions as intended before moving on to broader system assessments. In this phase, we systematically test core features—such as user signup and login flows, service listing creation, search and filtering, booking procedures, and messaging interfaces—by running predefined test cases that cover both expected behaviors and edge conditions. Automated unit tests validate the smallest units of code in isolation, while integration tests verify that related modules communicate correctly. Any failures are logged, investigated, and resolved, guaranteeing that all discrete components adhere to the requirements specified in earlier chapters.

6.1.1 Functionality Testing:

Functionality testing exercises the platform's main user-facing and administrative features under realistic scenarios. Test scripts simulate actions like creating a new service, searching across categories, completing a booking with payment, and exchanging messages in real time. Each test confirms that inputs produce the correct outputs, that error states are handled gracefully, and that workflows proceed smoothly from start to finish. Manual exploratory testing complements the automated suites, allowing testers to uncover unexpected behaviors or usability issues.

6.1.2 Static Testing

Static testing employs tools and processes that analyze code and configurations without executing the application. Linters check for syntax errors and enforce consistent style rules, while formatters automatically correct indentation and spacing. Type-checking tools catch mismatched data types and missing interfaces before runtime. Additionally, security scanners review dependency manifests to flag known vulnerabilities, and audit scripts verify compliance with internal coding standards. This proactive approach catches many potential bugs early in the development cycle, reducing downstream defects.

6.2. Validation

Validation in the SkillFlow project is the process of confirming that the developed system meets the intended use and aligns with the expectations of the end users, as defined during the requirements gathering phase. Unlike verification, which ensures that each component works as designed, validation ensures that the **right system has been built**—one that truly addresses the needs of freelancers and clients seeking to connect through the platform. This phase involves reviewing the system's real-world functionality in terms of reliability, usability, and user satisfaction. Various test scenarios are executed under realistic conditions to validate complete workflows such as user registration, service listing, project booking, and messaging features. Real user feedback is collected and analyzed to assess how intuitive and functional the platform is in practice. This feedback loop is critical for identifying any unmet needs or workflow obstacles that may not have been evident during development. Validation ensures that the final version of SkillFlow delivers a seamless, trustworthy, and value-driven experience, ultimately confirming its readiness for public release and adoption. Real user feedback is collected and analyzed to assess how intuitive and functional the platform is in practice.

Through validation, stakeholders can be confident that SkillFlow not only functions correctly but also delivers genuine value to its target users. By focusing on the platform's performance in real-life scenarios, validation bridges the gap between technical completion and user acceptance. It highlights whether workflows feel natural, interfaces are user-friendly, and features actually solve the problems freelancers and clients face in the freelance marketplace. This user-centered check ensures that the final product is not just technically sound, but also practical, engaging, and ready to achieve its intended.

Test Case ID	TC-001
Test Case Name	**Choose a role – User** **Create Account - Email Format Validation**
Description	Verify that valid email addresses are accepted for user registration.
Prerequisite	Database connection active.
Steps	• Open SkillFlow Website. • Navigate to Sign Up. • Enter valid email and password. • Afterwards, they clicked on "SIGN UP".
Expected Result	User will be successfully Registered.
Results	User is successfully Registered.
Pass/Fail	Pass

Table 6.1: Test Case for Sign-up with Valid Information

Test Case ID	TC-002
Test Case Name	**Create Account - Invalid Email Format Validation**
Description	Verify that the email format correctly identifies invalid email addresses.
Prerequisite	Connection with database established properly.
Steps	• Open SkillFlow. • The user clicked on "Sign Up". • Enter invalid email and password. • Afterwards, they clicked on "SIGN UP".
Expected Result	User will not be able to Register.
Results	User is not Registered.
Pass/Fail	Pass

Table 6.2: Test Case for Sign-up with Invalid Information

Test Case ID	TC-003
Test Case Name	**Login Account**

Description	Users must log into their account so they can see services provided by the application
Prerequisite	• Connection with database established properly. • Account is created by user.
Steps	• Application will login after inserting correct email and password. • Successfully login on correct information. • Easy access of account to user after successful login.
Expected Result	Successful login
Results	• Successfully login on inserting correct information. • Failed login issue on invalid information.
Estimated Time	• Successfully login within 15 seconds. • Also depends on internet speed.
Error	• Wrong email or password. • Account not created. • Internet connection error.

Table 6.3: Test Case for Log-in

Test Case ID	**TC-004**
Test Case Name	**Search Product**
Description	Validate correct search results for entered product names.
Prerequisite	• Database populated with product data.
Steps	Open Application Search product View results
Expected Result	Relevant product details displayed.
Results	Pass

Table 6.4: Test Case for Medicine Search

Test Case ID	**TC-005**
Test Case Name	**Provide Feedback**
Description	Users can submit feedback about the prediction results, and they are able to view their feedback data for each prediction. After the prediction, the recommendation screen is displayed.
Prerequisite	User must be logged in and have performed a prediction activity.
Steps	Admin views feedback data submitted by users. User views and provides feedback data for the prediction. Product recommendation screen is displayed after prediction.
Expected Result	To monitor and improve the prediction accuracy, and provide users with relevant product recommendations.
Results	Users successfully submit feedback for each prediction. Admin can view the feedback data provided by users.
Estimated Time	• Feedback submission within 15 seconds, subject to internet speed.
Error	• Admin fails to view feedback data due to a slow internet connection. • User fails to provide feedback due to a slow internet connection. • Internet connectivity issues.

Table 6.5: Test Case for Provide Feedback

Test Case ID	**TC-006**
Test Case Name	**Admin Authority-Add and Delete Product**
Description	Admin has the authority to add new entities and delete existing ones.
Prerequisite	Admin must be logged in and have the necessary permissions.
Steps	Admin accesses the management section. Admin selects the option to add a new entity. Admin fills in the required details and submits to add. Admin selects the option to delete an existing entity. Admin chooses the entity to delete and confirms the action.

Expected Result	Admin successfully adds a new entity and deletes an existing one.
Results	New entity added successfully. Existing entity deleted successfully
Estimated Time	• Adding a new entity: 1 minute. • Deleting an existing entity: 30 seconds. (Times may vary based on server response and internet speed.)
Error	• Admin encounters an error while adding due to incomplete information or server issues. • Deletion fails due to connectivity issues or unauthorized access.

Table 6.6: Test Case for Admin Authority - Add and Delete

6.3 Evaluation

The evaluation phase of the **SkillFlow** project was a structured and thorough review of all testing results to assess the system's overall quality, stability, and readiness for deployment. This stage involved analyzing data gathered during various testing activities—such as functional testing, integration testing, performance evaluation, and user acceptance testing—to ensure that the platform fulfilled both functional and non-functional requirements. The team examined reported defects, validated bug fixes, and ensured no critical regressions had occurred after updates or enhancements.

Evaluation sessions were conducted collaboratively, bringing together developers, QA engineers, UI/UX designers, and key stakeholders to discuss system performance, usability, and overall reliability. Key performance metrics like server response time, load handling capability, and error rates were reviewed against established benchmarks. Feedback from user testing—particularly regarding ease of use, navigation, and platform responsiveness—was closely studied to confirm that user-centric improvements had been effectively implemented.

Security assessments were revisited to verify that identified vulnerabilities were resolved and that the platform adhered to secure coding practices. Usability factors, such as clear interface elements, responsive design, and intuitive user flows, were also reevaluated to ensure a smooth experience across devices and user types.

Success criteria for the SkillFlow system included high pass rates across test cases, stable performance under moderate to high load, low error occurrences, and positive feedback from test users. Minor issues that did not impact core functionalities were documented for future maintenance cycles. The team also compiled detailed reports, feedback logs, and system health summaries for stakeholder review and sign-off. Through this comprehensive evaluation, the SkillFlow team confirmed the platform's readiness for live deployment and long-term use in a real-world freelance services environment.

6.4 Deployment

The deployment phase of the **SkillFlow** project was executed through a well-structured and controlled process to ensure a smooth transition from the development environment to a stable, accessible live platform. This stage included finalizing the application's source code, ensuring all components—frontend, backend, and database—were properly packaged, and that necessary configurations such as environment variables, authentication keys, and third-party service credentials were securely managed.

To streamline the release, automated deployment scripts and CI/CD pipelines were employed, reducing manual intervention and promoting consistency across development, staging, and production environments. These scripts facilitated code builds, database connections, asset compilation, and service restarts with minimal downtime. Prior to going live, final verification steps were conducted, including smoke testing and system-level checks, to confirm that essential features like user registration, service browsing, and project bookings functioned correctly in the production environment.

Database migrations were carefully managed to preserve existing data integrity, with backup routines in place to prevent accidental loss. A clear rollback mechanism was implemented, enabling the team to quickly restore a previous working state in the event of unforeseen deployment issues. Coordination among developers, testers, and stakeholders was essential, ensuring that all team members were informed of the deployment timeline, potential risks, and support protocols. This collaborative and proactive approach ensured that the SkillFlow platform was launched in a stable, secure, and fully operational state for real-world use.

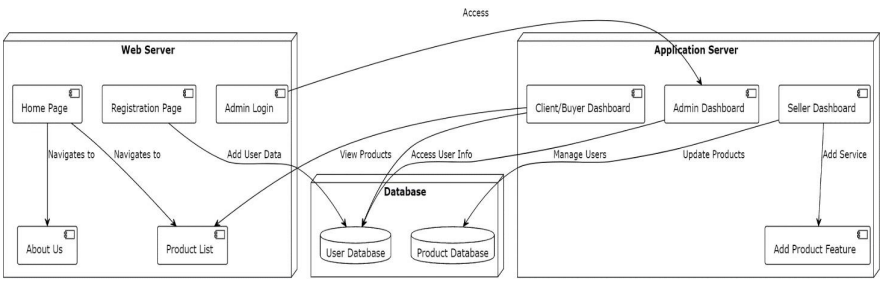

Fig 6.4: Deployment Diagram

6.5 Maintenance

The maintenance phase of the **SkillFlow** project marks the beginning of an ongoing effort to ensure the platform remains functional, secure, and adaptive after its deployment. Once the system was live, the development team immediately activated monitoring tools to observe platform behavior, track performance indicators, and capture real-time logs related to errors and user interactions. These tools help the team detect potential issues early, offering a proactive approach to system health and minimizing the impact of technical problems.

To safeguard against data loss, automated backup mechanisms were configured to regularly store critical information such as user profiles, service listings, booking histories, and communication logs. These backups are stored securely and are readily recoverable in the event of failures or accidental data corruption. Security maintenance plays a crucial role during this phase—routine updates are applied to both the system code and its external dependencies to address known vulnerabilities and ensure that SkillFlow is protected against evolving threats.

The maintenance team also maintains thorough and up-to-date documentation, including technical references for developers and support guidelines for user-facing teams. This ensures clarity when troubleshooting issues or introducing new features. Regular evaluation meetings are conducted to review bug reports, system performance analytics, and user feedback, which guide the prioritization of updates, fixes, and improvements.

Chapter 7: Conclusion and Future Work

7.1. Maintenance

The SkillFlow platform was developed to address a critical gap in the digital freelancing ecosystem, particularly for non-technical individuals in Pakistan. With a focus on empowering local communities, especially women and small-scale artisans, SkillFlow provides a dedicated, user-friendly, and secure platform to connect buyers and sellers of skill-based services. Unlike traditional freelancing platforms that cater mainly to technical domains, SkillFlow bridges the divide by offering services in non-technical areas like tailoring, baking, writing, and home-based crafts.

The development journey followed Agile methodology, ensuring iterative progress with continuous stakeholder feedback. Using a modern MERN (MongoDB, Express, React, Node.js) stack, the system delivers a responsive and scalable solution that supports real-time interactions, secure transactions, and personalized recommendations. The modular architecture and middleware integration ensured maintainability, scalability, and robust performance.

Through comprehensive requirement analysis, detailed design, efficient implementation, and rigorous testing, the platform was validated to meet both functional and non-functional requirements. From authentication to real-time messaging and payment integration, each component was developed with a strong focus on usability, performance, and security.

The successful deployment and evaluation phases confirmed SkillFlow's readiness for real-world usage and highlighted its potential to foster financial independence and local economic development.

7.2. Future Work

While SkillFlow has achieved its core objectives, several enhancements and expansions are planned to further increase its value and impact. From authentication to real-time messaging and payment integration, each component was developed with a strong focus on usability, performance, and security.

Future work includes:

7.2.1. Mobile Application Development

To improve accessibility, especially in rural areas with limited desktop access, a dedicated mobile application (Android/iOS) will be developed using React Native or Flutter.

7.2.2. AI-Based Recommendation System

Incorporating machine learning algorithms to enhance service discovery by analyzing user behavior, preferences, and feedback for smarter, more accurate recommendations.

7.2.3. Multilingual Support

Adding regional language support (Urdu, Punjabi, Pashto, etc.)

7.2.4. Region Based Promotions

Introducing a feature that allows buyers and sellers to interact based on region.

In conclusion, SkillFlow has laid a strong foundation for an inclusive and scalable online earning ecosystem. With thoughtful enhancements and a user-centric roadmap, the platform has the potential to evolve into a national and even regional hub for non-technical freelancing.

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